
Mitigating Reward Hacking in Discrete Diffusion Fine-Tuning for 5'UTR Design

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Abstract

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5 1 Introduction

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7 1.1 Style

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The formatting instructions contained in these style files are summarized in Sections ??, ??, and ?? below.

2 Preliminary

2.1 Discrete diffusion modeling sequence distribution

We present a 5’UTR sequence as $x = (x^{(1)}, \dots, x^{(L)})$ over the token alphabet $\mathcal{V} = \{\text{A, C, G, U, PAD}\}$, where L denotes the maximum sequence length. For a sequence with effective length $l < L$, positions $l + 1, \dots, L$ are filled with PAD tokens. We denote $\mathcal{X} = \mathcal{V}^L$ for the design space, the set of all-length- L token sequences over $\mathcal{V}$, and aim to learn a sampling distribution $p(\cdot) \in \Delta(\mathcal{X})$ over this space from the observed data. We adopt the masked discrete diffusion model (), a framework that has shown strong empirical performance in biology sequence generation (). The framework operates over an extended alphabet $\bar{\mathcal{X}} = (\mathcal{V} \cup \{\text{MASK}\})^L$, where the MASK is an auxiliary masking token. The principle of the framework includes: 1) a forward noising process that progressively corrupts clean data sequences into a noise distribution concentrated on the fully-masked state, and 2) a reverse denoising process that inverts this corruption to generate clean samples from noise. Both processes are formulated as continuous-time Markov chains (CTMCs) over $\bar{\mathcal{X}}$.

Forward Process. Given a clean sequence $x_0 \sim p_{\text{data}}$, the forward process independently replaces each token with MASK according to a predefined time-dependent schedule. For a single token position, let $p_t \in \Delta(\bar{\mathcal{V}})$ denote the marginal distribution over per-token states at time t , where $\bar{\mathcal{V}} = \mathcal{V} \cup \{\text{MASK}\}$. The forward CTMC is controlled by a transition rate matrix $Q(t) \in \mathbb{R}^{\bar{\mathcal{V}} \times \bar{\mathcal{V}}}$ via

$$\frac{dp_t}{dt} = Q(t)p_t, \quad p_0 = p_{\text{data}}, \quad (1)$$

where $Q(t)$ has nonzero entries only for data-token-to-MASK transitions. Concretely, this induces the per-position marginal

$$q(x_t^{(i)} | x_0^{(i)}) = \begin{cases} 1 - \sigma(t) & \text{if } x_t^{(i)} = x_0^{(i)}, \\ \sigma(t) & \text{if } x_t^{(i)} = \text{MASK}, \end{cases} \quad (2)$$

where $\sigma(t)$ is a monotonically increasing schedule with $\sigma(0) = 0$ and $\sigma(T) = 1$. At $t = T$, every position is masked with probability one, so the terminal distribution concentrates on the fully-masked sequence (MASK, ..., MASK) regardless of x_0 .

Reverse Process. The reverse process is itself a CTMC with generator $\bar{Q}^\theta(t)$, whose entries are parameterized by a neural denoiser $\hat{x}_0^\theta(x_t, t)$ that takes the full partially-masked sequence $x_t \in \bar{\mathcal{X}}$ as input and predicts, at each masked position, a distribution over clean tokens in $\mathcal{V}$. For a single token position, the reverse generator admits the closed form ()

$$\bar{Q}_{u,v}^\theta(t) = \begin{cases} \gamma(t) \cdot \hat{x}_0^\theta(v | x_t, t) & \text{if } u = \text{MASK}, v \neq \text{MASK}, \\ -\gamma(t) & \text{if } u = v = \text{MASK}, \\ 0 & \text{otherwise,} \end{cases} \quad (3)$$

for per-token states $u, v \in \bar{\mathcal{V}}$, where $\hat{x}_0^\theta(\cdot | x_t, t)$ denotes the denoiser’s predicted clean-token distribution at the position in question, and $\gamma(t)$ is a scalar derived from the forward schedule $\sigma(t)$. Intuitively, masked positions are unmasked at rate $\gamma(t)$ into data tokens drawn from $\hat{x}_0^\theta$, while already-unmasked positions are held fixed (their rows are zero). For sequences of length L , the sequence-level reverse generator factorizes across positions as a sum of per-position single-token generators (), avoiding exponential blowup in the state space.

2.2 Reward-guided fine-tuning via DRAKES

After the pretraining stage, we denote the pre-trained diffusion model as $p^{\text{pre}}(x) := p^{\theta_{\text{pre}}}(x)$, which serves as the starting policy for the reward-guided finetuning procedure. Our goal is to find a policy p^θ that can produce samples that maximize a specific reward value and keep the natural-like properties. Formally, we formulate this objective function as:

$$\arg \max_{\theta} \mathbb{E}_{x_0 \sim p^\theta} [r(x_0)] - \alpha \text{KL}(p^\theta \| p^{\theta_{\text{pre}}}), \quad (4)$$

where $r : \mathcal{X} \rightarrow \mathbb{R}$ is a reward function mapping sequences to task-specific scores. The first term encourages the fine-tuned policy p^θ to generate sequences with high reward, while the second term $\text{KL}(p^\theta \| p^{\theta_{\text{pre}}})$ penalizes deviation from the pretrained model $p^{\theta_{\text{pre}}}$, preserving the naturalness of generated sequences. The coefficient $\alpha > 0$ controls the trade-off between reward maximization and faithfulness to the pretrained distribution. To solve the reinforcement learning problem, policy gradient methods such as PPO (Schulman et al., 2017) serve as a natural solution. However, applying PPO to discrete diffusion finetuning is challenging: the long, non-differentiable CTMC sampling trajectory yields high-variance gradient estimates and can be susceptible to mode collapse during optimization (Su et al., 2025). We therefore adopt direct reward backpropagation with Gumbel-Softmax (DRAKES) proposed in (Wang et al., 2024), which renders the sampling trajectory differentiable and optimizes 4 via direct backpropagation. Specifically, DRAKES reformulates the objective as a trajectory-level expectation, with the marginal KL replaced by a KL between trajectory distributions induced by the reverse-time CTMC:

$$\begin{aligned} \theta^* = \arg \max_{\theta \in \Theta} \mathbb{E}_{x_{0:T} \sim p^\theta} [r(x_T)] \\ - \alpha \mathbb{E}_{x_{0:T} \sim p^\theta} \left[\int_{t=0}^T \sum_{i=1}^L \sum_{\substack{v \in \mathcal{V} \\ v \neq x_t^{(i)}}} \left\{ Q_{x_t^{(i)},v}^{\theta_{\text{pre}}}(t) - Q_{x_t^{(i)},v}^\theta(t) + Q_{x_t^{(i)},v}^\theta(t) \log \frac{Q_{x_t^{(i)},v}^\theta(t)}{Q_{x_t^{(i)},v}^{\theta_{\text{pre}}}(t)} \right\} dt \right], \end{aligned} \quad (5)$$

To make the objective differentiable in θ , DRAKES replaces the categorical sampling at each reverse step with the Gumbel-Softmax relaxation, producing a continuous surrogate trajectory $\bar{x}_{0:T}$, a straight-through estimator is applied at the final step so that the reward function $r(\cdot)$ receives a one-hot input while gradients still flow through $\bar{x}_T$. The empirical loss

$$\mathcal{L}_{\text{DRAKES}}(\theta) = \frac{1}{B} \sum_{i=1}^B \left[r(\bar{x}_T^{(i)}) - \alpha \widehat{\text{KL}}^{(i)}(\theta) \right] \quad (6)$$

is then optimized by stochastic gradient ascent, with gradients truncated at an intermediate reverse step. The detailed information can be referred to (Wang et al., 2024).

3 Reward hacking in 5'UTR sequence optimization

Despite its simplicity, the DRAKES objective in Eq. 5 is vulnerable to a well-known failure mode in RL fine-tuning. The fine-tuned policy may drift to out-of-distribution sequences on which $\hat{r}_\phi$ is poorly calibrated, exploiting predictive errors of the proxy reward rather than improving the true reward. The KL regularizer constrains how far the policy may drift from p^{pre} , but does not by itself resolve this miscalibration; the result is reward hacking—a rapid rise in $\hat{r}_\phi$ accompanied by degradation in the biological plausibility of generated sequences. To better understand the above challenges, we adopt DRAKES on the eGFP MPRA library of (Sample et al., 2019), comprising approximately 280,000 5'UTR sequences with measured MRL labels. We pretrain $p^{\theta_{\text{pre}}}$ on this corpus and train $\hat{r}_\phi$ via MSE regression using the corresponding labels, then fine-tune $p^{\theta_{\text{pre}}}$ with DRAKES across three KL strengths $\alpha \in \{0.001, 0.005, 0.01\}$. Figure ?? reveals a consistent pattern: under weak or moderate α , $\hat{r}_\phi$ converges rapidly to high values, while the 3-mer Pearson correlation between generated and natural high-MRL sequences collapses simultaneously. Stronger regularization caps the achievable reward without restoring naturalness, so the trade-off curve traced by varying α never reaches a regime where both reward and plausibility are high. Reward hacking in 5'UTR design is thus not a tuning issue but a structural failure of the vanilla RL finetuning.

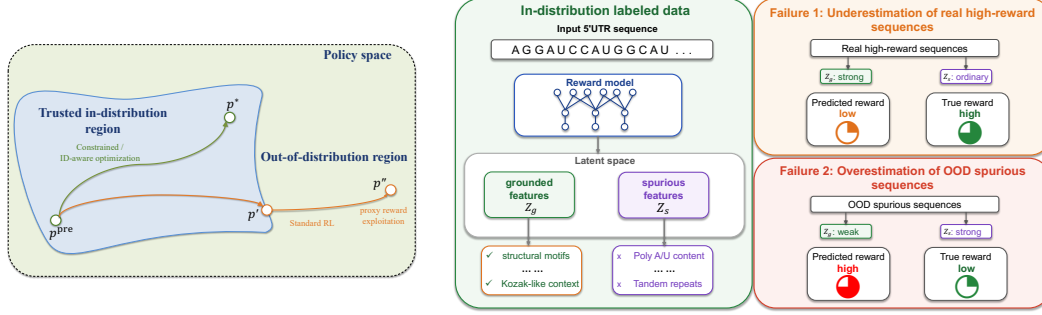


Figure 1: Motivation of calibrated and robust reward-guided fine-tuning for biological sequence design. (a) Standard RL may move the policy toward out-of-distribution regions where the proxy reward is exploitable, while constrained in-distribution optimization encourages movement toward reliable high-reward policies. (b) Reward models trained on in-distribution labeled data may fail under distribution shift by underestimating real high-reward sequences or overestimating OOD spurious sequences.

110 The remainder of this section diagnoses this failure. We trace the hacking dynamics to two character-
 111 istic miscalibrations of $\hat{r}_\phi$ (Section ??), compounded by the mode-seeking behavior of the reverse
 112 KL regularizer (Section ??)—a decomposition that motivates the three-component method developed
 113 in Section ??.

114 3.1 Failures of the learned reward model

115 **Failure 1: Underestimation of real high-reward sequences.** In the labeled 5'UTR dataset, se-
 116 quences in the upper tail of the MRL distribution—the practical target of reward-guided design—are
 117 difficult to fit accurately under a standard regression objective, as the loss in Eq. ?? is dominated by
 118 the bulk of moderate-score sequences in $\mathcal{D}_{\text{label}}$. Empirically, we observe on a held-out validation set
 119 that the top 10% of sequences by measured MRL (mean MRL ≈ 7.0) are predicted by the vanilla
 120 reward model at mean ≈ 5.8 , despite an overall Pearson correlation of approximately 0.7. This
 121 systematic underestimation suppresses the reward signal precisely in the region that reward-guided
 122 fine-tuning aims to reach, weakening the gradient toward genuinely high-reward sequences.

123 **Failure 2: Overestimation of out-of-distribution spurious sequences.** The second failure originates
 124 from shortcut learning on the in-distribution dataset $\mathcal{D}_{\text{label}}$. Following the latent feature perspective
 125 of (??), we decompose the learned representation of a reward model into two kinds of features
 126 (Figure ??): *spurious features* $z_s \in \mathcal{Z}_s$, which correlate with reward on $\mathcal{D}_{\text{label}}$ without causally
 127 determining it—for instance, natural 5'UTRs exhibit characteristic nucleotide-frequency patterns that
 128 correlate with, but do not cause, high MRL—and *functionally grounded features* $z_g \in \mathcal{Z}_g$, which
 129 are genuinely tied to translation efficiency, such as Kozak-like sequence contexts, the absence of
 130 upstream open reading frames, and appropriate GC composition surrounding the start codon. An
 131 ideal reward model depends only on z_g :

$$r^*(x) = f(z_g(x)). \quad (7)$$

132 In practice, however, the labeled dataset contains spurious correlations between z_s and MRL. For
 133 example, in the 50-nt random UTR library of (Sample et al., 2019), the A/U content of top-decile
 134 sequences is noticeably higher than that of the overall distribution, even though no causal mechanism
 135 links nucleotide composition to translation efficiency in this regime. A reward model fitted on such
 136 data therefore depends on both feature types,

$$\hat{r}(x) = \hat{f}(z_g(x), z_s(x)), \quad (8)$$

137 exhibiting the shortcut-learning pathology. The consequence becomes acute off-distribution: consider
 138 a sequence x_{OOD} with strong spurious attributes but negligible z_g content—precisely the kind of
 139 sequence a policy produces when it drifts off the training manifold during RL fine-tuning. Because $\hat{r}$
 140 has learned to associate high z_s values with high reward on $\mathcal{D}_{\text{label}}$, it extrapolates this association and

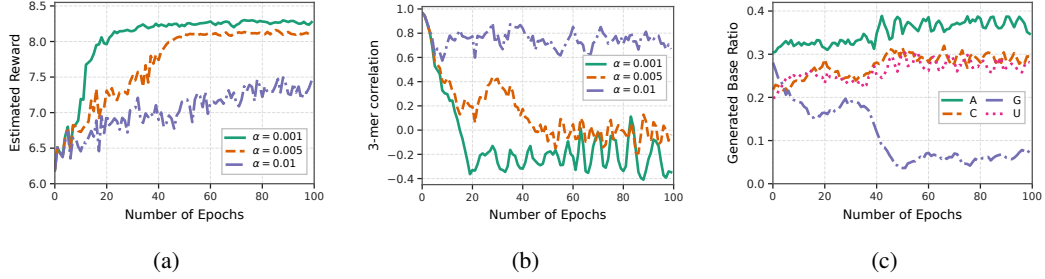


Figure 2: Illustration of reward hacking during RL fine-tuning. (a) Estimated reward trajectory. (b) 3-mer correlation trajectory. (c) Generated base-ratio trajectory.

141 assigns inflated predictions:

$$\hat{r}(x_{\text{OOD}}) \gg r^*(x_{\text{OOD}}). \quad (9)$$

142 Under the reward-maximization objective in Eq. ??, the policy is thereby drawn toward exactly
 143 this OOD spurious region, producing sequences whose *predicted* reward is high but whose *true*
 144 translational performance is not—the characteristic signature of reward overoptimization.

145 3.2 Failure of the reverse KL

146 Beyond the miscalibration of $\hat{r}_\phi$, the failure of vanilla DRAKES is compounded by a structural
 147 property of its reverse KL regularizer: *mode-seeking behavior* (Wang et al., 2023). When a policy is
 148 optimized to minimize $\text{KL}(p^\theta \| p^{\text{pre}})$, the optimum tends to concentrate on a sub-region of p^{pre} rather
 149 than covering it broadly, as shown in the Fig.?.?. This behavior is a consequence of the asymmetry of
 150 the KL divergence:

$$\text{KL}(p^\theta \| p^{\text{pre}}) = \mathbb{E}_{x \sim p^\theta} \left[\log \frac{p^\theta(x)}{p^{\text{pre}}(x)} \right], \quad (10)$$

151 which penalizes p^θ heavily wherever it places mass on regions with low p^{pre} , but is insensitive to
 152 regions where p^{pre} has high mass and p^θ has none. Minimizing this term therefore drives p^θ to
 153 concentrate within sub-regions of p^{pre} ’s support, with no incentive to cover the rest. This mode-
 154 seeking collapse manifests directly in our 5’UTR optimization with vanilla DRAKES. Figure ??
 155 tracks per-nucleotide composition over the course of fine-tuning under the moderate reverse KL
 156 constraint ($\alpha = 0.005$): as optimization proceeds, the policy progressively shifts toward A/U-rich
 157 sequences, drives the G ratio to near zero, and collapses onto a narrow sub-region of p^{pre} ’s support.
 158 As shown in Fig 2(a), tuning α alone cannot resolve this collapse: smaller α permits unrestricted
 159 hacking, while larger α stalls reward optimization at suboptimal levels, with no setting recovering
 160 both high reward and naturalness.

161 4 CARE: Calibrated and robust reward finetuning

162 We propose **CARE** (Calibrated And Robust rEward finetuning), a fine-tuning framework for discrete
 163 diffusion policies that mitigates reward hacking by addressing the two failure sources identified
 164 in Section ?? at their respective levels. To correct the miscalibrations of $\hat{r}_\phi$ (Section ??), CARE
 165 trains the reward model with a tail-reweighted regression loss to suppress underprediction on high-
 166 MRL sequences (Failure 1), and augments training with adversarial hacked samples and a cap loss
 167 to suppress inflated predictions on out-of-distribution inputs (Failure 2). To counteract the mode-
 168 seeking collapse of reverse-KL regularization (Section ??), CARE augments the reverse KL with a
 169 mass-covering forward KL term, forming a bidirectional regularizer that resists both directions of
 170 distributional drift.

171 4.1 Calibrated and Robust Reward Model Training

172 We motivate CARE’s reward-model training procedure by considering how the two failures of vanilla
 173 regression manifest at the level of individual sequences. Consider a high-MRL sequence x_{high}
 174 characterized by strong grounded features and weak spurious features, z_g^+ , z_s^- . Its true reward r_{high} is

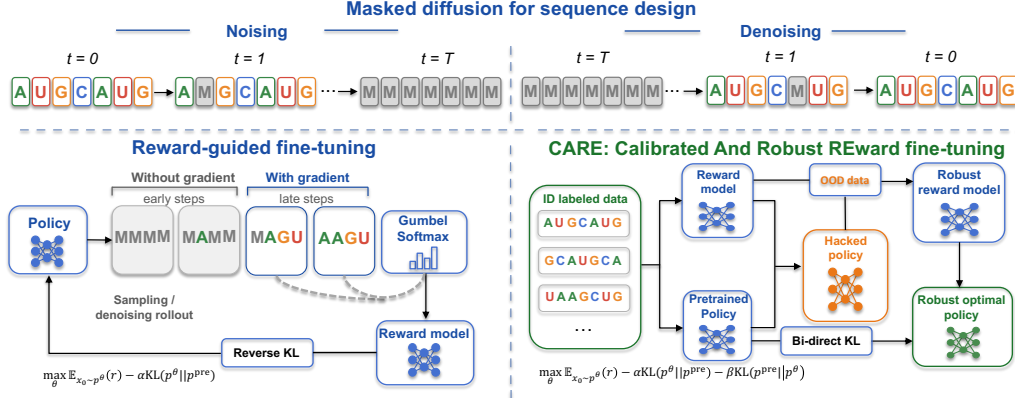


Figure 3: Overview of the proposed CARE framework for calibrated and robust reward-guided fine-tuning.

high, yet the vanilla reward model $\hat{r}_\phi$ systematically underpredicts it (Failure 1 of Fig.??). Conversely, consider an out-of-distribution sequence x_{OOD} characterized by strong spurious features and weak grounded features, z_g^-, z_s^+ . Its true reward is low—such sequences lack the structural motifs that drive translation—yet $\hat{r}_\phi(x_{\text{OOD}})$ is inflated by the model’s reliance on spurious features (Failure 2 of Fig.??). Our goal is to train a calibrated and robust reward model $\bar{r}_\phi$ that simultaneously corrects both failures: tightening predictions on high-reward sequences toward their true labels, while suppressing spurious extrapolation on OOD inputs.

Training objective. We formulate this as the joint minimization of two complementary losses:

$$\mathcal{L}_{\text{reward}}(\phi) = \underbrace{\mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{label}}} \left[w(y) \cdot (\bar{r}_\phi(x) - y)^2 \right]}_{\text{tail-reweighted regression}} + \lambda \cdot \underbrace{\mathbb{E}_{x_{\text{OOD}} \sim \mathcal{D}_{\text{adv}}} [\mathcal{L}_{\text{cap}}(x_{\text{OOD}})]}_{\text{adversarial cap loss}}, \quad (11)$$

where the first term re-emphasizes regression on high-reward sequences in $\mathcal{D}_{\text{label}}$, the second penalizes inflated predictions on adversarially-generated OOD sequences in $\mathcal{D}_{\text{adv}}$, and $\lambda > 0$ balances the two.

Tail-reweighted regression. To correct the systematic underprediction of high-reward sequences, we re-weight the regression loss by a function $w(y)$ that emphasizes samples in the upper tail of the label distribution and de-emphasizes samples in the lower tail. This redirects the regression objective away from being dominated by the bulk of low- and moderate-score samples, focusing it instead on accurate fitting in the upper tail.

Adversarial cap loss. To suppress inflated predictions on out-of-distribution sequences with strong spurious features (z_s^+), we augment the reward model training with adversarial samples and a cap loss.

How do we obtain such OOD sequences in the first place?

We construct the adversarial set $\mathcal{D}_{\text{adv}}$ via *self-distillation*: we fine-tune the pretrained policy $p^{\theta_{\text{pre}}}$ against the vanilla reward model $\hat{r}_\phi$ under a deliberately weak KL coefficient, allowing the policy to drift into the OOD hacking region characterized in Fig. ?? . Sampling from the resulting policy p'' yields sequences with inflated predicted reward driven by strong spurious features—precisely the inputs that the calibrated reward model $\bar{r}_\phi$ must learn to refuse. Given a sequence x_{OOD} from $\mathcal{D}_{\text{adv}}$, the cap loss takes the form of a one-sided hinge penalty:

$$\mathcal{L}_{\text{cap}}(x_{\text{OOD}}) = \max(0, \bar{r}_\phi(x_{\text{OOD}}) - \sigma), \quad (12)$$

where σ is a threshold above which $\bar{r}_\phi$ ’s predictions on OOD sequences are deemed implausibly high. We set σ to the median of the training reward distribution in $\mathcal{D}_{\text{label}}$. This choice reflects the assumption that an OOD sequence with no genuinely grounded features should not be predicted

Table 1: Model performance on 5'UTR sequence optimization across MRL and TE datasets. For each dataset, we report reward optimization performance using the mean predicted reward and the average predicted reward of the top 100 sequences selected from 1000 generated candidates. We evaluate naturalness using 3-mer correlation with the validation set and with the top 10% high-reward validation sequences. We further report the average minimum free energy (MFE) as a biophysical property. Results are averaged over 3 random seeds, with standard deviations in parentheses.

2*Method	MRL Optimization					TE Optimization				
	Mean↑	Top-100↑	3-mer Val↑	3-mer Top10↑	MFE↓	Mean↑	Top-100↑	3-mer Val↑	3-mer Top10↑	MFE↓
Pretrained	-	-	-	-	-	-	-	-	-	-
CG	-	-	-	-	-	-	-	-	-	-
SMC	-	-	-	-	-	-	-	-	-	-
TDS	-	-	-	-	-	-	-	-	-	-
CFG	-	-	-	-	-	-	-	-	-	-
UTRGAN	-	-	-	-	-	-	-	-	-	-
DRAKES	-	-	-	-	-	-	-	-	-	-
CARE	-	-	-	-	-	-	-	-	-	-

higher than a typical natural training sample. Together, the hinge penalty weakens the dependence of $\bar{r}_\phi$ on spurious features z_s and shifts its predictive support onto grounded features z_g .

4.2 Bidirectional KL regularization

Beyond a calibrated reward model, mitigating reward hacking requires a regularizer that prevents the fine-tuned policy from drifting away from p^{pre} . As discussed in Section ??, the standard reverse KL term is mode-seeking: it permits the policy to concentrate on narrow sub-regions of p^{pre} 's support without penalty, including the hacked region. We address this by introducing a complementary forward KL term, which is more conservative and distribution-preserving. Combining the two yields a bidirectional KL regularizer:

$$\theta^* = \arg \max_{\theta} \mathbb{E}_{x_0 \sim p^\theta} [\bar{r}_\phi(x_0)] - \alpha \text{KL}(p^\theta \| p^{\text{pre}}) - \beta \text{KL}(p^{\text{pre}} \| p^\theta), \quad (13)$$

where $\alpha > 0$ controls the strength of the mode-seeking reverse KL, and $\beta > 0$ controls the strength of the mass-covering forward KL. The two terms together resist both forms of distributional drift identified in Section ??.

5 Experiment

These instructions apply to everyone.

5.1 Citations within the text

The natbib package will be loaded for you by default. Citations may be author/year or numeric, as long as you maintain internal consistency. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

The documentation for natbib may be found at

<http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf>

Of note is the command `\citet`, which produces citations appropriate for use in inline text. For example,

```
\citet{hasselmo} investigated\dots
```

produces

Hasselmo, et al. (1995) investigated...

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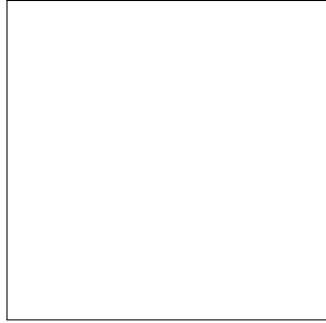


Figure 4: Sample figure caption. Explain what the figure shows and add a key take-away message to the caption.

230 `\PassOptionsToPackage{options}{natbib}`

231 If `natbib` clashes with another package you load, you can add the optional argument `nonatbib`
232 when loading the style file:

233 `\usepackage[nonatbib]{neurips_2026}`

234 As submission is double blind, refer to your own published work in the third person. That is, use “In
235 the previous work of Jones et al. [4],” not “In our previous work [4].” If you cite your other papers
236 that are not widely available (e.g., a journal paper under review), use anonymous author names in the
237 citation, e.g., an author of the form “A. Anonymous” and include a copy of the anonymized paper in
238 the supplementary material.

239 5.2 Footnotes

240 Footnotes should be used sparingly. If you do require a footnote, indicate footnotes with a number¹
241 in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote
242 with a horizontal rule of 2 inches (12 picas).

243 Note that footnotes are properly typeset *after* punctuation marks.²

244 5.3 Figures

245 All artwork must be neat, clean, and legible. Lines should be dark enough for reproduction purposes.
246 The figure number and caption always appear after the figure. Place one line space before the figure
247 caption and one line space after the figure. The figure caption should be lower case (except for the
248 first word and proper nouns); figures are numbered consecutively.

249 You may use color figures. However, it is best for the figure captions and the paper body to be legible
250 if the paper is printed in either black/white or in color.

251 5.4 Tables

252 All tables must be centered, neat, clean, and legible. The table number and title always appear before
253 the table. See Table 2.

254 Place one line space before the table title, one line space after the table title, and one line space after
255 the table. The table title must be lower case (except for the first word and proper nouns); tables are
256 numbered consecutively.

257 Note that publication-quality tables *do not contain vertical rules*. We strongly suggest the use of the
258 `booktabs` package, which allows for typesetting high-quality, professional tables:

259 <https://www.ctan.org/pkg/booktabs>

¹Sample of the first footnote.

²As in this example.

Table 2: Sample table caption. Explain what the table shows and add a key take-away message to the caption.

Part		
Name	Description	Size (μm)
Dendrite	Input terminal	≈ 100
Axon	Output terminal	≈ 10
Soma	Cell body	up to 10^6

260 This package was used to typeset Table 2.

261 5.5 Math

262 Note that display math in bare TeX commands will not create correct line numbers for sub-
 263 mission. Please use LaTeX (or AMSTeX) commands for unnumbered display math. (You
 264 really shouldn't be using \$\$ anyway; see <https://tex.stackexchange.com/questions/503/why-is-preferable-to> and [https://tex.stackexchange.com/questions/40492/](https://tex.stackexchange.com/questions/40492/what-are-the-differences-between-align-equation-and-displaymath)
 265 [what-are-the-differences-between-align-equation-and-displaymath](https://tex.stackexchange.com/questions/40492/what-are-the-differences-between-align-equation-and-displaymath) for more infor-
 266 mation.)

268 5.6 Final instructions

269 Do not change any aspects of the formatting parameters in the style files. In particular, do not modify
 270 the width or length of the rectangle the text should fit into, and do not change font sizes. Please note
 271 that pages should be numbered.

272 6 Preparing PDF files

273 Please prepare submission files with paper size "US Letter," and not, for example, "A4."

274 Fonts were the main cause of problems in the past years. Your PDF file must only contain Type 1 or
 275 Embedded TrueType fonts. Here are a few instructions to achieve this.

- 276 • You should directly generate PDF files using `pdflatex`.
- 277 • You can check which fonts a PDF files uses. In Acrobat Reader, select the menu
 278 Files>Document Properties>Fonts and select Show All Fonts. You can also use the program
 279 `pdf fonts` which comes with `xpdf` and is available out-of-the-box on most Linux machines.
- 280 • `xfig` "patterned" shapes are implemented with bitmap fonts. Use "solid" shapes instead.
- 281 • The `\bbold` package almost always uses bitmap fonts. You should use the equivalent AMS
 282 Fonts:

283 `\usepackage{amsfonts}`

284 followed by, e.g., `\mathbb{R}`, `\mathbb{N}`, or `\mathbb{C}` for $\mathbb{R}$, $\mathbb{N}$ or $\mathbb{C}$. You can also
 285 use the following workaround for reals, natural and complex:

```
286 \newcommand{\RR}{\mathbb{R}} %real numbers
287 \newcommand{\Nat}{\mathbb{N}} %natural numbers
288 \newcommand{\CC}{\mathbb{C}} %complex numbers
```

289 Note that `amsfonts` is automatically loaded by the `amssymb` package.

290 If your file contains type 3 fonts or non embedded TrueType fonts, we will ask you to fix it.

291 6.1 Margins in L^AT_EX

292 Most of the margin problems come from figures positioned by hand using `\special` or other
 293 commands. We suggest using the command `\includegraphics` from the `graphicx` package.
 294 Always specify the figure width as a multiple of the line width as in the example below:

295 `\usepackage[pdftex]{graphicx} ...`
 296 `\includegraphics[width=0.8\linewidth]{myfile.pdf}`

297 See Section 4.4 in the graphics bundle documentation ([http://mirrors.ctan.org/macros/](http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf)
 298 [latex/required/graphics/grfguide.pdf](http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf))

299 A number of width problems arise when L^AT_EX cannot properly hyphenate a line. Please give LaTeX
 300 hyphenation hints using the `\-` command when necessary.

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318 A Technical appendices and supplementary material

- 319 Technical appendices with additional results, figures, graphs, and proofs may be submitted with the
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